

COGS 160 · A0 Article Checklist

Q1: CO2 Concentration & Decision Quality (K-ATLAS Q16) · Dhruv Sood (d2sood@ucsd.edu)

Research Question (Q16): At what CO2 concentration does strategic decision-making quality (measured by SMS or comparable instrument) begin to decline, and is the threshold lower for cognitively demanding vs. routine tasks?

Submission Checklist

✓	10 empirical papers identified	DONE
✓	All 10 are empirical (original data, human subjects, CO2 as IV or measured exposure)	DONE
✓	All 10 PDFs downloaded (open access sources)	DONE
✓	APA 7 citations complete (apa_citations.txt)	DONE
✓	Manifest CSV complete — all fields filled (title, authors, year, doi, abstract, source, notes)	DONE
✓	All DOIs verified as resolvable	DONE
✓	Search queries documented (queries.md)	DONE
■	Q2 PDFs: 5/10 downloaded; 5 still need UCSD library access	PENDING

10 Papers — PDF Inventory & Relevance to Q16

All PDFs confirmed as real (not HTML errors). Sources: Europe PMC, IZA, eScholarship, npj, ERL, MDPI.

#	Reference & PDF File	Study Design & Outcome	Key Finding & Relevance to Q16
1	Satish et al. (2012). Environmental Health Perspectives, 120(12), 1671-1677. doi:10.1289/ehp.1104789 ✓ satish_2012_co2_decision.pdf OA: Europe PMC (PMC3548274)	Design: Double-blind RCT chamber study 22 adult participants 3 CO2 levels (600, 1000, 2500 ppm) Single day, 3x 2.5-hr sessions Measure: SMS (Strategic Management Simulation) — 9 subscales incl. strategic planning, crisis response, info usage	Finding: At 1000 ppm: significant decrements on 6/9 SMS scales vs. 600 ppm baseline. At 2500 ppm: large reductions on 7/9 scales (score ratios 0.06–0.56). Focused activity scale unaffected — first evidence of task-type threshold difference. Q16 link: DIRECTLY answers Q16. Establishes 1000 ppm as the key threshold; provides SMS evidence that demanding tasks decline before routine ones.
2	Allen et al. (2016). Environmental Health Perspectives, 124(6), 805-812. doi:10.1289/ehp.1510037 ✓ allen_2016_co2_cognition.pdf OA: Europe PMC (PMC4892924)	Design: Controlled exposure, within-subjects 24 office workers 6 full work days Green / Conventional / Green+ IEQ conditions; CO2 independently manipulated Measure: SMS-based cognitive scores across 9 domains (higher-order cognitive function battery)	Finding: Cognitive scores 61% higher on Green day, 101% higher on Green+ days vs. Conventional (p<0.0001). CO2 and VOCs independently associated — CO2 effect isolated from ventilation confound. Q16 link: Replicates Satish 2012 in realistic office simulation; isolates CO2 from VOC and ventilation; shows effect extends across ALL 9 cognitive domains.
3	Scully et al. (2019). npj Microgravity, 5, 17. doi:10.1038/s41526-019-0071-6 ✓ scully_2019_co2_cognition.pdf OA: npj Microgravity (open access, Nature Publishing)	Design: Astronaut-analog crossover 0 to 1.2% CO2 exposures SMS + NASA Cognition test battery Simulated spacecraft environment Measure: SMS decision-making scores + 10 computerized cognitive tests (processing speed, attention, WM, executive function)	Finding: No clear monotonic dose-response found. Performance decrements at 1200 ppm but recovery at higher concentrations — suggests population differences (astronaut analogs vs. office workers) and multi-stressor interactions. Q16 link: Tests Q16 with the SMS instrument in extreme conditions; raises nuance — threshold may vary by population/stressor context; uses both SMS and objective cognitive battery.

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4	Cedeno Laurent et al. (2021). Environmental Research Letters, 16(9), 094047. doi:10.1088/1748-9326/ac1bd8 ✓ cedeno_2021_co2_pm25_cognitive.pdf OA: IOP Publishing (CC-BY); PMC8942432	Design: Prospective observational longitudinal 302 office workers 6 countries 12 months Real-time CO₂ + PM_{2.5} monitoring Mobile app cognitive tests Measure: Stroop color-word test (response time, interference, throughput) + Addition-Subtraction test throughput — administered daily via smartphone	Finding: IQR CO₂ increase (315 ppm) associated with +0.85% Stroop RT, +7.88% interference time, -1.32% throughput. PM_{2.5} effects also significant. Generalizes across 6 countries. Q16 link: Largest real-world sample for CO₂-cognition link; validates Satish 2012 findings outside the lab; directly tests demanding (Stroop interference) vs. routine (RT) task difference.
5	MacNaughton et al. (2015). Int. J. Environ. Res. Public Health, 12(11), 14709-14722. doi:10.3390/ijerph121114709 ✓ macnaughton_2015_ventilation_cognitive.pdf OA: MDPI IJERPH (gold OA); PMC4661675	Design: Economic modeling study Energy cost estimation for 7 U.S. cities (different climate zones) Ventilation scenarios: 20, 30, 40 cfm/person Uses SMS productivity data from Satish 2012 Measure: Per-person annual energy cost vs. SMS-derived worker productivity gain across ventilation levels	Finding: Doubling ventilation (halving CO₂) costs <\$40/person/year in all climates. Same change improves SMS-measured performance by 8% = \$6500/yr productivity gain. Health benefits far exceed energy costs. Q16 link: Bridges Q16 to real-world policy — quantifies the economic value of keeping CO₂ below 1000 ppm threshold; applies SMS productivity data at scale across building stock.
6	Vehvilainen et al. (2016). J. Occupational & Environmental Hygiene, 13(1), 19-29. doi:10.1080/15459624.2015.1076160 ✓ vehvilainen_2016_co2.pdf OA: eScholarship repository (open deposit)	Design: Repeated-measures physiological study Male volunteers 4-hr simulated office sessions Ventilated vs. non-ventilated conditions Physiological monitoring throughout Measure: Transcutaneous CO₂ level, heart rate, peripheral blood flow, subjective sleepiness (KSS) during cognitive work tasks	Finding: High CO₂ (non-ventilated) significantly increased transcutaneous CO₂, altered cardiovascular parameters, and increased sleepiness. Physiological effects provide a mechanism by which elevated CO₂ impairs cognitive work capacity. Q16 link: Identifies physiological mechanism linking elevated CO₂ to cognitive impairment — fills mechanistic gap in Q16 by showing CO₂ directly affects the body, not just air quality perception.
7	Kunn, Palacios, & Pestel (2019). IZA Discussion Paper No. 12632. doi:10.2139/ssrn.3460848 ✓ kunn_2019_iaq_cognitive.pdf OA: IZA Discussion Papers (docs.iza.org, free)	Design: Quasi-experimental observational 30 professional chess tournaments Chess engine analysis of move quality PM_{2.5} and CO₂ measured at tournament venues Causal identification via within-player variation Measure: Probability of making a suboptimal move (error rate) vs. IAQ conditions — chess move quality as a naturalistic high-stakes decision-making proxy	Finding: 10 ug/m³ PM_{2.5} increase raises error probability by 26.3%. CO₂ effects concentrated during time-pressured phases (fast decision-making). Suggests demanding, time-constrained decisions are most affected. Q16 link: Natural experiment with 30 tournaments answers Q16 with a real-world high-stakes decision task; directly tests cognitively demanding (time pressure) vs. routine phases.
8	Shimazaki et al. (2025). Environmental and Occupational Health Practice, 7(1). doi:10.1539/eohp.2025-0001 ✓ shimazaki_2025_co2_driving_taxi.pdf OA: PMC12461243 (open access)	Design: Simulator experiment 8 professional taxi drivers Driving license proficiency criteria <500 ppm vs. 5000 ppm CO₂ Poisson + GLMM analysis Measure: Driving errors (wobble, signal failure, contact, wheel departure) + LED reaction time + 2-back working memory task	Finding: High CO₂ (5000 ppm) significantly increased all driving error types (p<0.05). LED reaction time significantly slower (p<0.001). 2-back WM task: no significant effect. Skill-based behavior impaired more than WM. Q16 link: Directly addresses Q16 threshold comparison — driving is a complex real-world decision task; results confirm cognitively demanding skill-based decisions impaired > routine memory task.
9	Young et al. (2024). Building and Environment, 261, 111551. doi:10.1016/j.buildenv.2024.111551 ✓ young_2024_co2_wfh_cognition.pdf OA: PMC11221786 (open access)	Design: Longitudinal observational 206 U.S. remote workers 1 year (2021-2022) Consumer IAQ monitors + smartphone app Stroop, ADD, cRAT cognitive tests geofenced to home Measure: Stroop interference (demanding), arithmetic throughput (moderate), Compound Remote Associates Task — creative problem-solving (demanding)	Finding: CO₂ mostly <640 ppm in homes; slight but significant association between higher CO₂ and Stroop performance (p=0.09). Thermal conditions showed non-linear effects. Highlights IAQ matters even at sub-1000 ppm levels. Q16 link: Extends Q16 evidence to home/WFH settings at lower CO₂ concentrations; suggests demanding cognitive tasks (Stroop interference, cRAT) may show effects even below the 1000 ppm Satish threshold.

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10	Bejder Klausen et al. (2023). Int. J. Occup. Med. Environ. Health, 36(2), 177-191. doi:10.13075/ijomeh.1896.02032 ✓ bejder_klausen_2023_co2_sleep_cognitive.pdf OA: PMC10464806 (open access)	Design: Double-blind, placebo-controlled, fully balanced crossover RCT Children ages 10-12 CO2 manipulated during sleep Next-morning cognitive performance measured Strong causal design Measure: Next-morning cognitive performance battery — attention, processing speed, executive function tasks	Finding: CO2 exposure during sleep significantly impacted next-morning cognitive performance. Crossover design controls for individual differences. Demonstrates that CO2 effects extend beyond waking hours — sleep-phase exposure carries over to daytime decision capacity. Q16 link: Adds novel dimension to Q16 — CO2 can degrade decision-relevant cognition via sleep-quality pathway, not just direct daytime exposure; RCT provides strongest causal evidence in the set.

Files on disk: ~/cogs160/A0_collect_articles/Q1_assigned/pdfs/ (10 PDFs) · manifest.csv · apa_citations.txt · queries.md
GitHub: https://github.com/dhruvsood12/COGS_160 (commit 88b0004) · Zip: ~/cogs160/A0_collect_articles.zip (13 MB)
Q2 (Circadian Lighting x Cognition): 5/10 PDFs present — remaining 5 available via UCSD library proxy